

Design Brief

1. Introduction

BA1130 is a classroom in the Bahen Centre for Information Technology located in the University of Toronto. The room hosts hours of lecture daily, including first year Engineering Science students for their Engineering Mathematics and Computation course along with Computer Programming. It is the professor's service to provide education and notes to the student, and a blackboard is often the main method of communication used by many professors in order to convey the course curriculum to the students when writing things like equations, diagrams, or proofs. However, due to the lack of better alternatives, professors continue to use the board even though it is not an efficient medium of communicating information. The current board in BA1130 has two superposed black boards, and their position on the wall can be changed by pushing the board up by a set of grooves attached to the walls on either side of the board. The board is able to stay in its position without moving due to hanging over a hidden pulley, and the use of counterweights [1].

The two sources of immediate cleanup within a lecture is the ineffective use of a chalkboard eraser and the side of the human hand. The use of a blackboard may hinder a professor in how they teach their lectures as chalk dust quickly accumulates and it becomes more and more difficult or time-consuming to clean up. Other professors try a different approach in lecture such as the use of projectors, but in their approach fail to use the full potential of the space provided to them by the classroom. They instead sacrifice on other subtle aspects of learning provided by the use of a chalkboard such as failing to respond to and illustrate a student's question in real time on the board for the whole class to see. They can annotate information and label graphs, but it becomes tedious and time consuming for the professor to do so ahead of time through powerpoint.

There are many requirements that define a solution, considering different metrics, constraints, and stakeholders. The following design report will outline these requirements which will provide enough information for interpretation to address the opportunity. Although this design brief specifically focuses on BA1130, its metrics, constraints, and stakeholders can apply to other blackboards on campus. As a pivotal education tool used in the University of Toronto, improving its functionality would positively impact the school.

2. Issues with the current setup of the blackboards

2.1. Impairment of presenting information

2.1.1. Visibility of blackboards

The repeated use of the chalkboard renders it increasingly difficult to read and understand due to the overlaid layers of chalk dust, smudging anything written afterwards, and due to the damage that board accumulates over time (scratches, cracks, fading color). Students viewing the board may have difficulty reading what is being written by the professor because the color of the

board is actually approaching that of the professor's chalk itself. This adds an extra step to the students' otherwise streamlined note-taking process in that they have to interpret and distinguish what is written, thus slowing them down. As shown in Appendix A1, the contrast ratio of the dirty board to writing in chalk does not pass WCAG accessibility guidelines and the City of Toronto Accessibility Design Guidelines requires may visual elements of Toronto to be "highly visible [with] contrasting colors" and does not allow background colors that provide poor contrast such as grey.

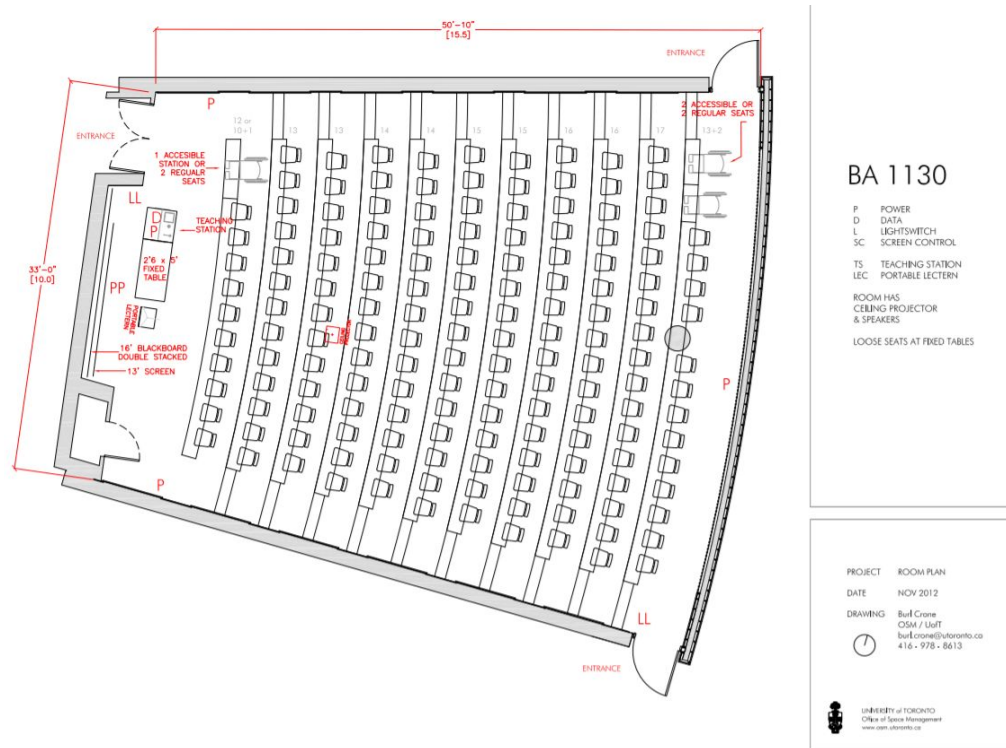


Figure 1: Floor plan of BA1130.

2.1.2. Multi-media presentations not suitable for blackboards

The use of tools like projectors give an additional audio and visual aspect to a lesson. But in order to actually use a projector a screen must pulled down over the board, thereby blocking it and making it functionless for the duration of the lecture while the screen is down. As shown in **Figure 1**, there is a small section of the blackboard that is 3 feet in width available to use but it does not allow the professor to give a lecture using both the projector and the board. And if the screen were to break then until it is fixed the blackboard would have no way to present different types of media.

2.2. Blackboards are more difficult for some to use

2.2.1 Health hazard of chalk dust

Although the main ingredients of chalk are non-toxic, constant and prolonged exposure can lead to long-term health problems [3]. The first type of chalk dust exposure is through the respiratory system. When blackboard erasers become oversaturated, they do a poor job of picking up all of the dust particles, and instead they begin to release these particles into the surrounding air of the classroom or lecture hall. This leads to respiratory problems such as coughing as well as lacrimation [4]. Experts recommend wearing filtered nasal and mouth masks, as well as taking breaks between use of the board [3]. Professors, on the other hand, must talk during their use of a chalkboard due to their reliance on verbal communication throughout the lecture which may last for prolonged periods of time. The second method of chalk dust exposure is through the skin. Professors use erasers to remove the chalk writing from a large area of the board, but often times resort to using the side of their hand as a quick method to erase a small mistake. This causes a saturated patch of chalk on one's hand for the period of the lecture which affects the blood stream since the skin begins to absorb the chalk dust [3]. This affects the rest of their organs, and as a result, the professor's teaching abilities are negatively impacted.

2.2.2 Inaccessibility of blackboards

In order to optimize the vertical space of the wall at the front of the room, two boards are superposed, allowing for more information to be placed overhead. The second board allows for students to copy the notes while the professor is writing on the other one. It also allows the students sitting in higher rows to easily see the board that is now level with their position. The blackboard system requires the professor to push up and pull down the boards, which is not possible for professors in wheelchairs as their maximum reach, 1370 mm as shown in **Figure 2**, is less than the height of the upper board.

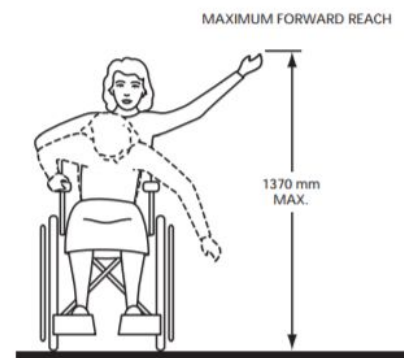


Figure 2: Maximum reach of a person in a wheelchair [8].

3. Stakeholders

- Professor
 - May choose to have lessons on the blackboard
 - Need a way to effectively present information
- Students
 - The board should be a tool to facilitate learning for students
- Cleaning/maintenance staff
 - Need to easily be able to clean board

- Might remove entire board in case of a need for maintenance
- Environment
 - Need something that is durable and produces little waste

4. Engineering requirements

4.1 Objectives

- 4.1.1. Improve visibility so that students are able to see clearly no matter where they are seated [Appendix A1].
- 4.1.2. Allow for quick transition between writing and multimedia.
- 4.1.3. Improve the accessibility of the board, making it usable by anyone no matter what their personal conditions are (i.e. Wheelchairs [Appendix A1], allergies, ...).
- 4.1.4. Make the board easy to move (if required).
- 4.1.5. Quickly and easily clean and use board. [Appendix A3]
- 4.1.6. Limit chalk dust generation to reduce the possibility of health problems and soiled clothing.
- 4.1.7. Improve the durability of the board and its accompanying equipment (ex: retractable screen / consumables)[Appendix A2]
- 4.1.8. Develop easy to replace structure so that repair and maintenance can be made by simply replacing certain part of the board.[Appendix A3]
- 4.1.9. Lower the cost of consumptive parts of the system (ie. chalk, markers) and boards, as well as the installation cost.

4.2 Constraints

- 4.2.1. Must have written words display high contrast against the board [Appendix A1].
- 4.2.2. Must allow multi-media presentations.
- 4.2.3. Must fit in current size classrooms.
- 4.2.4. Should not waste available vertical space by leaving it empty.
- 4.2.5. Must have protection at the bottom in case the board falls accidentally.
- 4.2.6. Must have an option for professors to hand-write in real-time.

4.3 Criteria

- 4.3.1. Prefer writing space with most amount of writing space.
- 4.3.2. Prefer writing space that creates the least amount of chalk dust.
- 4.3.3. Prefer writing space with high durability of writing and erasing equipments.
- 4.3.4. Prefer writing space with quick clean up.
- 4.3.5. Prefer writing space that is able to present the most forms of information.

4.4 Metrics

- 4.4.1. Area of writing space.
- 4.4.2. Amount of chalk residue after use (if any).

4.4.3. Life span of the consumptive writing and erasing equipments [Appendix A2].

4.4.4. Time spent erasing content from the writing space.

4.4.5. Time spent switching between different types of presentation forms.

4.4.6. Color contrast between writing space and writing or text.

5. Reference designs

5.1 Whiteboard

- Markers are encased in plastic and are disposed of frequently if markers are used often (which they will be in a classroom with about 50 lectures a week), though there exist some types of refillable markers. [4.3.3.]
- Cannot endure long term use as board eventually becomes hard to wipe clean [5] [4.4.3.1.]
- Same problem with chalkboard eraser: eraser endures build up [4.3.4.]

5.2 Projector

- Classrooms are often provided with 1 screen, meaning there's limit space to show information. The professor must quickly move aside previous notes to make room for new ones. Students may not be able to take down everything. [4.3.1.]
- Professor stands in one position the entire lecture. May cause pain.
- Cannot use board behind it. [4.2.2.]

5.3 Smart Board

- Board is fixed; cannot use maximum amount of wall space because professors can't reach that high [4.2.4.]
- More expensive than blackboards

5.4 Glass

- Cannot project images or videos on it [4.2.2.]
- Hard to take a picture of notes on
- Words are harder to see [4.1.1.]

5.5 Auto clean eraser

- Too much noise [6]
- Slow to act. Takes more time than simply erasing. [6] [4.3.4.]

6. Conclusion

Through an analysis of the design of accessibility, durability, and maintainability of BA1130, a set of objectives, metrics, criteria, and constraints were set with the intention of providing enough information for a solution to be found. Alternatives were also analyzed and none satisfied all the objectives, criteria, and constraints. The issue with the current blackboards provides an opportunity to use engineering design to address it and develop a conceptual design. The design concept can then be applied to other blackboards on campus.

Appendix A: DfX Site Inspections

1. Accessibility

Designing for accessibility aims to help overcome the obstacles that are experienced by people who have disabilities, which may include hearing, mobility, and vision. Through a combination of field notes and first hand experience from sitting in lectures in the BA1130 auditorium, two problems were identified that make aspects of the board not accessible. There is a poor color contrast between a dusty blackboard and writing on the board, and the placement of the board is too high above the ground to be reachable to people in wheelchairs.

First, a prevailing issue with the use of the blackboard is that writing is hard to see when the board is covered with a layer of chalk dust. With the board covered in a shade of white, it becomes hard to read writing in white chalk, which is used by Professor Cluett and Mathai in BA1130. This creates even bigger problem for colorblind people, who have problems with color perception, and people with low vision.

The Web Content Accessibility (WCAG) is a set of color contrast compliances set by the World Wide Web Consortium (W3C). The guidelines aims to make content more accessible for people with visual impairments, such as color blindness and low vision, amongst other disabilities [7]. The WCAG AA for Large Text guideline requires that the contrast ratio of large text (minimum 18 point, which is smaller than the writing of a professor on the board) be at least 3:1, calculated with formulas they have defined.

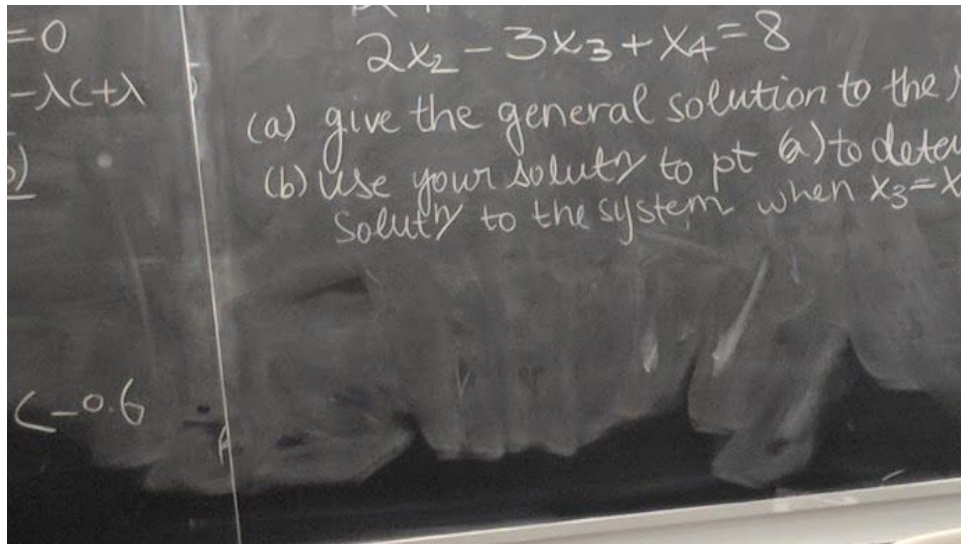


Figure 3: A blackboard used for ESC103 in BA.

Using a color picker to find the hexadecimal code of the colors, an image of a blackboard, **Figure 3**, was taken and three colors were analyzed, the color of the writing (#F7F4EF), the color of the board covered with dust (#9A9997), and the board without dust (#252525). When checking the constant ratio

between the writing and dusty board, it failed the WCAG AA for Large Text guideline, as well as all the other guidelines for different sized texts, with a contrast ratio of 2.59:1 [8]. The constant ratio between the writing and clean board passed all tests, with a ratio of 13.97:1 [8]. According to the WCAG 2.0 standards, writing on a dirty board fails accessibility guidelines, thereby making it difficult for students to read off of. The City of Toronto Accessibility Design Guidelines also says to use “highly visible and contrasting colors” and says that background colors such as grey are “unacceptable” [9].

Second, the board is not accessible for disabled people in wheelchairs. In an attempt to maximize the use of the wall vertically, the board is allowed to slide up along the wall. However, it is out of reach and cannot be used by wheelchair users.

According to the City of Toronto Accessibility Design Guidelines [9], the maximum reach of a person in a wheelchair is 1370 mm (54 inches). In BA1130, the upper board is approximately 2000 mm from the floor. This is high out of reach for people in wheelchairs and renders the second board useless for their situation.

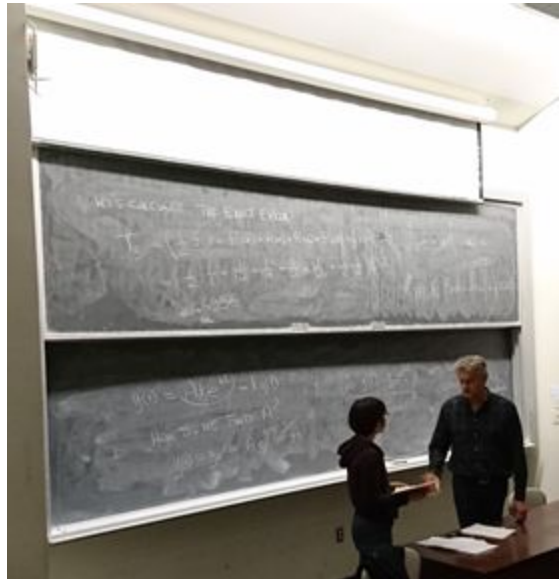
In conclusion, the design of the current blackboard is not accessible for the student and professor stakeholders. First, students have problem seeing the board due to the low contrast of the text from the board. As a result, this creates extra burden for students who are taking down notes. This opportunity is addressed in objectives 4.1.1. and 4.1.3. Second, the board is not ideal for users in wheelchairs, as they cannot reach the upper board. This is brought up in objective 4.1.2. Addressing these concerns should help improve the accessibility of the new design.

2. Durability

The blackboards that can currently be found in the classes of the Bahen Center for information technology, and the ones that are currently being used by professors and students alike are standard, chalk blackboards. The blackboard is an almost archaic class tool, one that many institutions have decided to replace, and for good reason. Some of these reasons have to do with it's durability, in fact with a quick look at the boards in BA it's possible to identify signs that show how old they are (i.e. faded color / scratches and markings). This site inspection will address the blackboard from a durability design perspective, it will talk about the materials used in the manufacturing of a board, the effects of wear on the board after years of use, and the different mechanisms and consumables that accompany the board.

The typical blackboard, including the ones in BA, are usually made from a steel sheet base, a porcelain enamel body, and a thin coat of inorganic compounds[10]. There are several thicknesses of steel used in the manufacturing of chalkboards, but the most common is the 22-gauge (0,0299 inches[11]), this aspect of the board mostly determines how easily the boards can shatter when subjected to bigger forces, like if

the board falls off of the wall and unto the floor, though that is one the less likely ways a board gets damaged. As for porcelain enamel, it's the main material of the outer layer of the black board and the part on which you can write. Also called vitreous enamel, it has many properties that make it suitable for being used in blackboards, it's smooth, hard, chemically resistant, and relatively scratch resistant (5-6 on the Mohs[12] scale) which means it won't be damaged by chalk. Though this doesn't mean it can't be damaged, it can still be scratched by materials with a higher Mohs scale and enamel has a tendency to crack or shatter when stressed. Lastly the coating on blackboards is a blend of inorganic materials that gives the board it's color while also making it smoother, this means that as the coat starts to wear from the constant use of the board, the board will also start to discolor.



When damaged these blackboards can't easily be repaired because, whether the damage is superficial or deep the process of repair would be long, expensive, and would have to be done in a specialized area, at that point it would be simpler to completely replace them.

In order to make use of the vertical space in a class, while also allowing professors to reach the higher parts of the boards, a mechanism is put in place in the class that allows two boards to switch vertical positions. This mechanism, that works thanks to series of pulleys hidden in the metallic edges of the board[1], when used improperly or when subjected to too much force, can also break (as seen in some classes in the PB building such as PB B250). When this happens, using the two boards can become very difficult which in turn makes teaching more difficult, it can also lead to one of the boards dropping which is both unsafe and damaging to the board.

The screen that is pulled down when a projector is used also suffers from similar issues. If the mechanism that allows the screen to be retractable were to be damaged from misuse, then the screen would become a burden, either stuck in its closed state and not allowing for projections, or stuck in it's open state and limiting the writing space on the board. Not only that but the cloth material that screen is made of is prone to shredding and burning.

Lastly, as for the consumables related to a blackboard, the main two are chalk and chalk erasers. A professor giving a lecture will go through one or two sticks of chalk per hour on average, this means that, on average, during a 9 hour day a blackboard requires around 13 pieces of chalk. The chalk eraser on the other hand while it can last a couple months of constant use if it's a good quality one, will still require daily cleaning in order to fully erase and not leave behind dust and discoloration.

The setup for the blackboards in BA suffer from multiple issues from a durability point of view, while it could hypothetically last for a dozen years if used properly and not subjected to any unnecessary damage it would still degrade over time becoming worse to use with the boards color fading, metal rusting, and cloth wearing down. And in the more realistic case where the setup is damaged, in some cases it could become unusable unless repaired or replaced. Not only that but constantly using a blackboard burns through chalk and chalk erasers, both of which create residue. Therefore a better option must be found that can last longer, that suffers more easily repairable damage, that has more re-usable consumables, and can be used just as easily no matter how old it is.

3. Maintainability

Design for maintainability aims to improve the duration of a clean writing space as lectures go on and also the easiness of operation process needed to repair the board. Through observation of a few blackboards in different lecture halls (mainly in BA1130), two problems were identified that make blackboards not well maintainable. Despite the presence of erasers, erasing the content fully from the board fails. Non-ergonomic blackboard structures cannot satisfy the needs of professors and lecturers, which results in inappropriate usage.

First, an issue with the use of the blackboard is that the currently provided erasers cannot fully erase content from the board, making it hard to maintain a clear contrast black surface to write on. During one lecture, professors have to frequently erase the board to write down more notes[13.1]. And the erasers get saturated too fast (mostly after the first use). Instead of doing their works of erasing chalk marks, they end up spreading out chalk dust among the whole board, in which case the boards lose their clear colour contrast[Appendix A1][13.2]. This slows down notes taking speed of students, since they have to stop from time to time to figure out what word it exactly is. This could also result in incorrectness of knowledge delivery, since some words could be mistaken[13.4].

The chalk with which we write on boards is gypsum, the dehydrated form of calcium sulphate, a loose mineral substance. It leaves light coloured small particles adhering to board surface due to friction between two rough surface. And chalk erasers are typically made of sponge or soft cloth, taking advantage of friction to remove chalk dust, small particles left on board. However, these materials cannot absorb dust and need regular cleaning to knock loose all that chalk adhere[15]. This result in spreading dust all over the board. And the background colour becomes grey, which is not “acceptable” [Appendix A1].

Secondly, design of mobile boards cannot satisfy the needs of professors and lecturers, resulting in inappropriate usage, which in some cases may damage the boards. In order to enlarge the writing space,

superposed boards are used. Their positions can be changed by pushing the board up by a set of grooves attached to the wall on both sides of the board. The boards are able to stay stably in their position only at the bottom or top position. They are able to stay relatively stable in the middle due to hanging over a hidden pulley with the use of counter mass, with no external forces acting on them. However, they fail to stay in their position while writing and sometimes fall down gradually. Also when pulled down heavily, there is no slow down reaction, causing heavy and unexpected load on the supports. Supports are gradually damaged and eventually fall. We witnessed one quick fall of blackboard and support in calculus class. This shrinks the lifespan of a blackboard.

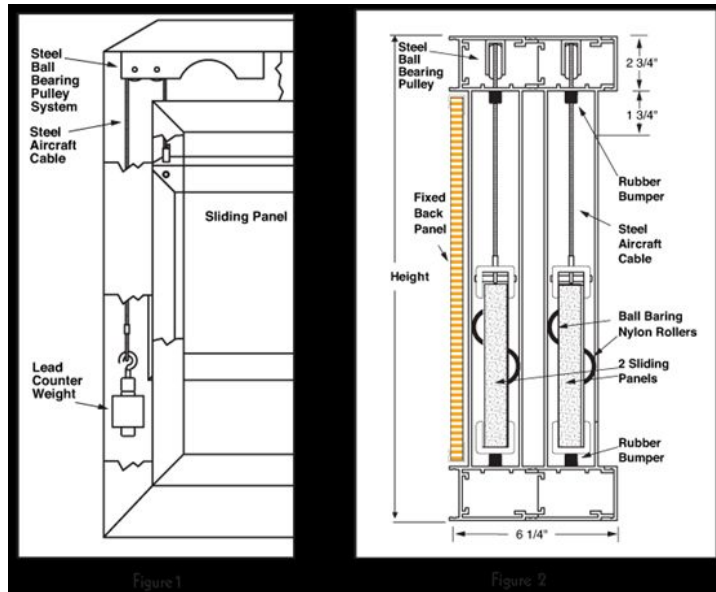
The blackboard is approximately 90 cm off the ground[13.4], which is a design satisfying the elevation constraint, 3'[16]. However, this constraint is used for any fixed position boards instead of superposed mobile board system. Professor Mathai reported that the boards cannot slide down to a lower height he wanted at which he could reach the top of the board. He also mentioned that blackboards have restricted motion below a certain point for some reason. Therefore, sometimes he pushes the boards too hard, which might cause permanent damage to them.

Last but not the least, the built-in structure of the blackboard disables regular check and repair access. All mechanisms are embedded in wall, with no simple direct access. Once a certain part fails, the whole structure has to be disassembled for repair

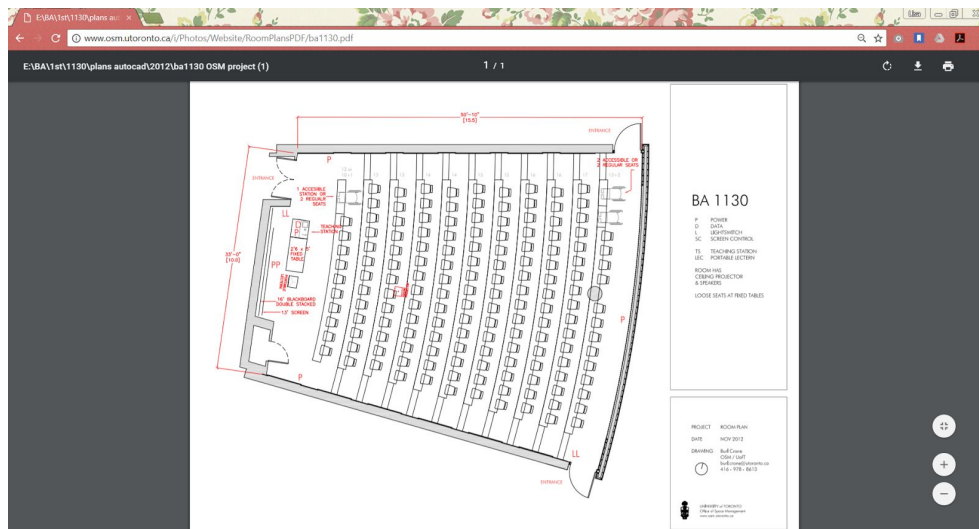
The blackboard system suffers from three main issues considering the maintainability. First, a clear writing space cannot be maintained during lectures. Second, a less considerate structure and simple mechanism fail to address the needs of user, causing inappropriate uses which shortens the lifespan. Built-in structure into the wall disabled the access to its mechanism, making them tedious to repair (maintenance workers have to fully disassemble the whole blackboard system in order to repair.

Source Extracts

[1]"Vertical Sliding Units", *Aarco.com*, 2017. [Online]. Available: <http://aarco.com/05.htm>



[2] "Classroom Descriptions", *Academic and Campus Events*, 2012. [Online]. Available: http://www.osm.utoronto.ca/class_spec/f?p=210:1:1278053919169001::::P1_BLDG,P1_ROOM:,: [Accessed: 01- Nov- 2017]



[3] Hassan, M.F. (2011). Investigation of Lecturer's Chalk by x-ray Florescence and Fast Neutron Activation Techniques. *Proceedings of the eighth Nuclear and Particle Physics Conference (NUPAC-2011)*, (p. 287). Egypt

“Our skins were thought of as an impenetrable barrier, but now we know about it differently. Substances that come into contact with the skin are absorbed and eventually find their way into the blood

stream. Toxins and other harmful products accumulate in the vital organs over a period of time causing many problems with our bodies. Lecturer's chalk is a native hydrous magnesium silicate sometimes containing a small portion of aluminum silicate [1] or calcium carbonate. It is similar in its chemical composition as asbestos, a well known cancer-causing agent. Chalk's dust harmful effect on human tissues has been known for quite some time [2-6].

There are two separate issues buried in the question of chalk dust safety. In one sense, the main ingredients of chalk dust are considered to be non-toxic, which simply means they do not pose a threat when ingested. In another sense, chalk dust can and does accumulate in the human respiratory system, which means it can create long-term health problems due to over exposure. In short, swallowing a piece of white chalkboard chalk won't kill you, but breathing in chalk dust for a number of years can create or trigger respiratory problems [7].”

[4] D. Majumdar, D. Gajghate, P. Pipalatkhar and C. Chalapati Rao, "Assessment of Airborne Fine Particulate Matter and Particle Size Distribution in Settled Chalk Dust during Writing and Dusting Exercises in a Classroom", *Indoor and Built Environment*, vol. 21, no. 4, pp. 541-551, 2011. [Accessed: 03- Nov- 2017].

"During teaching, entry of chalk dust in the respiratory system through nasopharyngeal region and mouth could be extensive in teachers due to their proximity to the board and frequent opening of mouth during lectures and occasional gasping and heavier breathing due to exhaustion. As per current state of knowledge on particulate matter vis-a-vis chalk dust, it "may remain suspended in air for some time before settling on the floor and body parts of the teachers and pupils".

"Though real-time airborne chalk dust generation was found to be low in this study ... and did not contain toxic materials, chalk dust could be harmful to allergic persons and may cause lacrimation and breathing troubles in the long run and certainly is a constant nuisance in classrooms as it may soil clothes, body parts, audiovisual aids and study materials."

[5] "Whiteboard Staining and Ghosting", *Glass Whiteboards and Glass Dry Erase Boards by Clarus*, 2017. [Online]. Available: <https://www.clarusglassboards.com/whiteboard-staining-and-ghosting/>. [Accessed: 01- Nov- 2017].

“While the dry erase marker typically wipes clean from surfaces, most whiteboards have a slight static charge to help the pigment stick to the surface. Over time whiteboards no longer wipe clean and can leave ugly stains and ghosted images. Once your board starts staining, you will need to clean it every few uses, to ensure a clean and streak-free board. Not only can it be a pain to constantly clean your boards, but the additional chemicals will add wear and tear and reduce your board’s lifespan.”

[6] *Auto-Clean Eraser: Demonstration that shows how loud and slow it is*. 2007. Available: <https://www.youtube.com/watch?v=1xwCwbKrZQc>.

[7] "Web Content Accessibility Guidelines (WCAG) 2.0", *The World Wide Web Consortium (W3C)*, 2017. [Online].

Available: <https://www.w3.org/TR/2008/REC-WCAG20-20081211/#visual-audio-contrast-contrast> [Accessed: 01- Nov- 2017].

“Web Content Accessibility Guidelines (WCAG) 2.0 covers a wide range of recommendations for making Web content more accessible. Following these guidelines will make content accessible to a wider range of people with disabilities, including blindness and low vision, deafness and hearing loss, learning disabilities, cognitive limitations, limited movement, speech disabilities, photosensitivity and combinations of these. Following these guidelines will also often make your Web content more usable to users in general.

“1.4.3 Contrast (Minimum): The visual presentation of text and images of text has a contrast ratio of at least 4.5:1, except for the following: (Level AA)

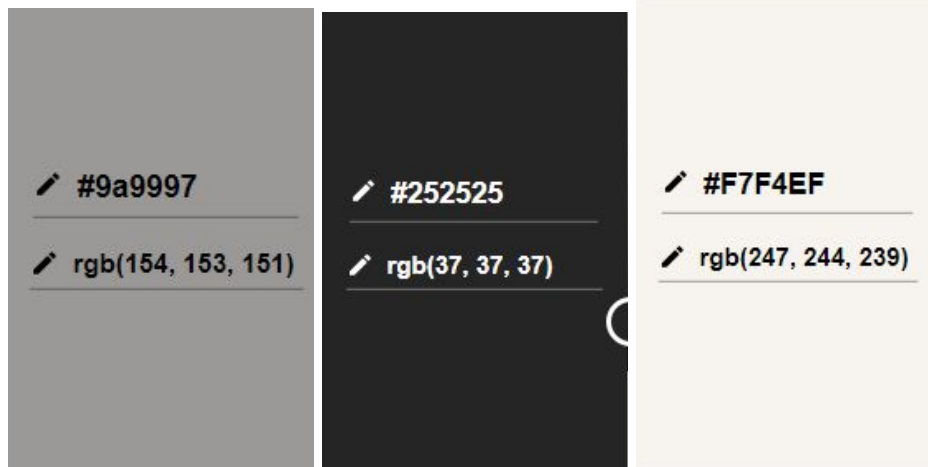
- **Large Text:** Large-scale text and images of large-scale text have a contrast ratio of at least 3:1;

...

large scale (text)

with at least 18 point or 14 point bold or font size that would yield equivalent size for Chinese, Japanese and Korean (CJK) fonts”

[8] "WebAIM: Color Contrast Checker", *Webaim.org*, 2017. [Online]. Available: <https://webaim.org/resources/contrastchecker/>. [Accessed: 01- Nov- 2017].



Color Contrast Checker

[Home](#) > [Resources](#) > Color Contrast Checker

Foreground Color

Lightness


Background Color

Lightness


Contrast Ratio
2.59:1
[permalink](#)

Normal Text

WCAG AA: **Fail**

WCAG AAA: **Fail**

The five boxing wizards jump quickly.

Large Text

WCAG AA: **Fail**

WCAG AAA: **Fail**

The five boxing wizards jump quickly.

[Feedback](#) [Help](#) [Privacy](#)

Color Contrast Checker

[Home](#) > [Resources](#) > Color Contrast Checker

Foreground Color

Lightness


Background Color

Lightness


Contrast Ratio
13.97:1
[permalink](#)

Normal Text

WCAG AA: **Pass**
WCAG AAA: **Pass**

The five boxing wizards jump quickly.

Large Text

WCAG AA: **Pass**
WCAG AAA: **Pass**

The five boxing wizards jump quickly.

[9] City of Toronto, "Accessibility Design Guidelines", Toronto, 2017.

“Signage should generally be designed using highly visible and contrasting colours (e.g., white or yellow on a black, charcoal, or other dark background such as brown, dark blue, dark green, or purple). Black lettering on white or yellow matte surfaces is also acceptable. Unacceptable background colours are: light grey or pastels. “

“Essential print information should generally be printed in large text (e.g., 12-14 pt bold) on a highly contrasting background colour. Print information should also be available in alternate formats, including braille or audiotape, for use by persons who have visual limitations.”

“All controls and operating mechanisms should be reachable and usable by persons with varying disabilities. (See Figures 130 and 131)”

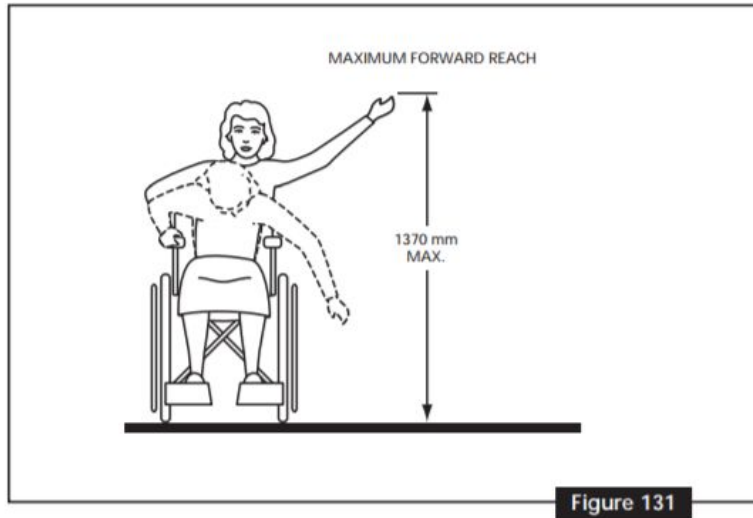


Figure 131

[10]"How chalkboard is made - material, manufacture, making, history, used, steps, product, Raw Materials", *Madehow.com*, 2017. [Online]. Available: <http://www.madehow.com/Volume-2/Chalkboard.html>.

"Most modern chalkboards are made of porcelain enamel. In this particular manufacturing process, a tough and durable material such as steel is used as the base. There are several thicknesses of steel used in the manufacture of chalkboards, but the most common is 22-gauge. Another crucial element is silica, a crystalline compound derived from quartz or similar minerals. Found in the crust of the earth, [silicon](#) is a tough compound and is called silica when combined with oxygen. Silica is found in most rocks and is a common ingredient in many glass and [ceramic](#) products. The surface of a chalkboard is usually a blend of inorganic compounds such as a powdered glass opacifier and oxides, an organic element that provides color to the coating material."

[11]"Sheet Metal Thickness Gauges", *Unc.edu*, 2017. [Online]. Available: <https://www.unc.edu/~rowlett/units/scales/sheetmetal.html>

Gauge (ga)	Standard Steel Thickness (inches)
3	0.2391
4	0.2242
5	0.2092
6	0.1943
7	0.1793
8	0.1644
9	0.1495
10	0.1345
11	0.1196
12	0.1046
13	0.0897
14	0.0747
15	0.0673
16	0.0598
17	0.0538
18	0.0478
19	0.0418
20	0.0359
21	0.0329
22	0.0299
23	0.0269
24	0.0239
25	0.0209
26	0.0179
27	0.0164
28	0.0149

[12]"Mohs scale of mineral hardness", *En.wikipedia.org*, 2017. [Online]. Available: https://en.wikipedia.org/wiki/Mohs_scale_of_mineral_hardness

“The Mohs scale of mineral hardness is a qualitative-ordinal scale characterizing scratch resistance of various minerals through the ability of harder material to scratch softer material”

[13]Field Notes

1. Lecturers have to frequently erase those boards to get space for more informations
2. After several times of use, the eraser gets saturated with chalk dust and spread the dust all over the board and worsen the visibility of contrast of writing. Note that janitors only clean the blackboard once a day. The boards' condition becomes worse and worse as day progresses. Boards are covered by chalk dust and approach the same colour as chalk.[figure 2]
3. Students frequently ask questions about what certain letter/number is.
4. Board is approximately 90 cm off the ground(54'' from floor plans)
5. Shadow blocks the top part of bottom board, slowing down the speed of notes taking. Students are having hard time figuring out the actual content.[figure 1]
6. Professor Mathai hates the blackboard because it cannot slide down to a height he wants, and he cannot reach the top of the bottom board. He sometimes pushes the board too hard, which is probably destroying the wheel track of the board.



Nebu Mathai <nebu.mathai@utoronto.ca>

Today, 8:20 PM

Lisa Li



Reply all

The blackboards have the following issues:
 Poor contrast ratio; especially after the board is used and erased
 Chalk doesn't hold an edge (doesn't stay sharp)
 Chalk is messy
 Chalk boards are less amenable to colour presentation

I would prefer white boards because they address the above.

The boards in the lecture halls seem to also cast a shadow on the lower boards, and have restricted motion below a certain point for some reason.

7. Board is approx 90 cm off the ground (36" from floorplans)
8. Bottom of top board is approx 200 cm of the ground
9. Width of board available with projector screen down is 3'
10. Shadow blocks top part of bottom board

[14] Y. Zhu. (2017). *Images taken of blackboards in PB B250*. Toronto, ON.

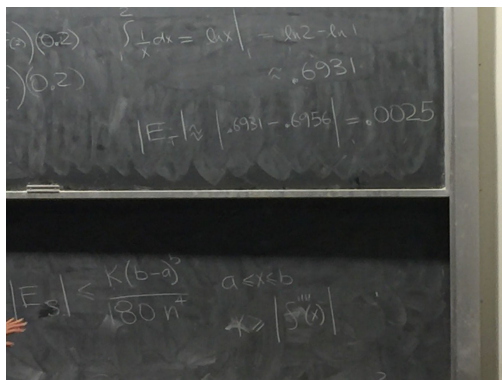


Figure 1

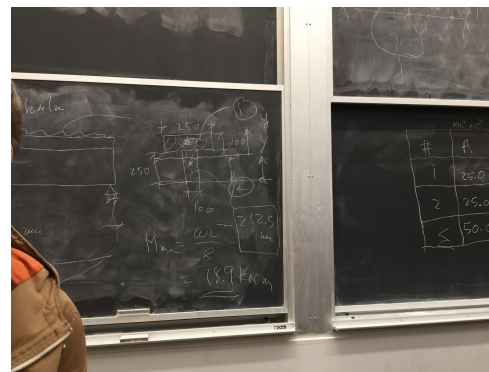


Figure 2



Figure 3

[15] Buzbee, L. (2017). *The Most Ingenious Teaching Device Ever Invented*. [online] Slate Magazine. Available at: http://www.slate.com/articles/life/education/2014/10/a_history_of_the_blackboard_how_the_blackboard_became_an_effective_and_ubiquitous.html [Accessed 4 Nov. 2017].

“Slates and chalk boards were often cleaned with dry rags, and no doubt sleeves, but in the late 19th century, erasers were developed for this task, blocks of wood (later pressed cardboard) covered with tufted felt, usually black or gray. These erasers needed regular cleaning to knock loose all that chalk crammed into the felt’s pores.”

“Chalk writing leaves light-coloured particles weakly adhering to a dark surface (e.g., white on black, or yellow on green); it can be rubbed off with a soft material, such as a rag. Erasers for chalkboards are made, with a block of plastic or wood, much larger than an eraser for pen or pencil, with a layer of felt on one side.”

[16]Jones, J. (1992). *Design methods*. New York: Van Nostrand Reinhold, p.141.